THE NUMERAL BASE SETS THE FOURIER FEATURES LANGUAGE MODELS COMPUTE WITH

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ABSTRACT

Language models represent numbers with periodic Fourier features whose periods, such as 2, 5, and 10, match the decimal system, and use them for arithmetic through a clock algorithm. Prior work identified these features by reading their periods from activations and intervening on them, but left open why models learn these particular periods. One explanation is the Chinese Remainder Theorem (CRT). Periods 2 and 5 are coprime and multiply to 10, so their residues together fix a decimal digit, which could favor periods that divide the base. Another is that the periods come from clocks that complete 1, 2, 3, 4, or 5 full cycles over ten digits and so have periods 10, 5, 10/3, 5/2, and 2, which we call the harmonic family of the base. Both accounts predict periods 2, 5, and 10, so observing them cannot distinguish the two, but only the harmonic account predicts periods such as 10/3 and 5/2. We therefore pretrain matched language models on text and arithmetic in bases 6, 7, 8, and 10. In every base, the model learns the harmonic family of its base, including periods 7/2 and 7/3 in base 7, where CRT favors only period 7. Interchange interventions on feature angles show that each harmonic separates the digits its period can distinguish. Period 5 groups digits five apart, period 2 separates even from odd, and in base 7 every harmonic gives each digit its own position. Near the output, the contributions of individual harmonics to digit scores add approximately in all four models and in Qwen3.5-9B. In earlier Qwen layers, however, patching all five harmonics together produces up to 2.0 times the sum of their individual effects for the units digit and up to 2.9 times for the tens digit, revealing nonlinear amplification before the final readout. We also trace how carry propagates through these features. Carrying one into the tens position shifts every tens-place harmonic toward the phase expected for the next digit. A small population of MLP neurons, under 1% of those in the upper layers, writes the units and tens clocks jointly, and replacing its activations transfers carry-dependent tens predictions between inputs, showing that interlocking clocks can implement carry without a separate carry bit.

1 INTRODUCTION

Prior work identified periodic number features in language models, with periods such as 2, 5, 10, and 100 (Zhou et al., 2024; Kantamneni & Tegmark, 2025; Fu et al., 2026), by reading them from activations and then intervening on them (Zhou et al., 2024; Kantamneni & Tegmark, 2025; Feucht et al., 2026). Models use these features for arithmetic through what is often called a clock algorithm (Nanda et al., 2023; Zhong et al., 2023; Kantamneni & Tegmark, 2025). These studies establish which features exist and that some are causal, but not why models learn these periods rather than others.

Two explanations could favor the observed periods. One is a residue account based on the Chinese Remainder Theorem (CRT). Because 2 and 5 are coprime and multiply to 10, features with these periods together fix the units digit, which could favor periods that divide the base. The other is a harmonic account. With the ten digits arranged around a clock, a feature can complete 1, 2, 3, 4, or 5 full cycles over the ten digits, giving periods 10, 5, 10/3, 5/2, and 2, which we call the *harmonic family* of base 10. Each clock has as many stops as the digits it can tell apart. Clocks that complete 1 or 3 cycles visit all ten stops, clocks that complete 2 or 4 cycles have five stops, so digits five apart share one, and the clock that completes 5 cycles has two stops, even and odd (Figure 1). Patching

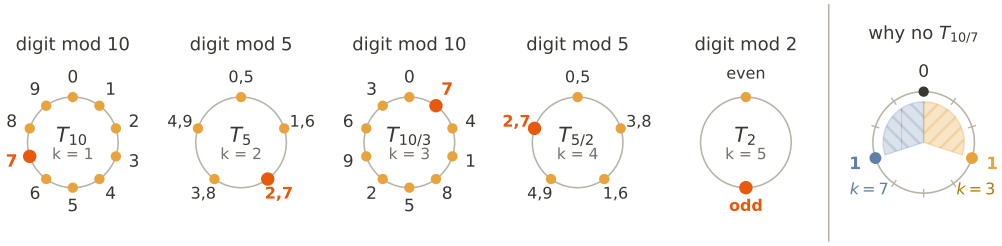


Figure 1: The base-10 units family as clocks with ten positions. Harmonic k moves k positions per digit. Periods 10 and $10/3$ give every digit its own position, periods 5 and $5/2$ place digits five apart together, and period 2 places all even digits at one position and all odd digits at the other. Orange marks digit 7 and the digits that share its position. Right, harmonic 7 moves seven positions per digit, which is three positions backwards, so it repeats harmonic 3 in reverse and adds no new feature.

a clock raises every digit at the target's stop by the same amount. A clock that completes 7 cycles moves three positions backwards per digit and spans the same plane as harmonic 3, adding no new information (Figure 1, right). Because this family contains every period that divides 10, observing periods 2, 5, and 10 cannot distinguish the two accounts. They differ on periods that do not divide the base, such as 10/3 and 5/2, which only the harmonic account predicts. To separate them, we pretrain matched 126M-parameter models on text and arithmetic in bases 6, 7, 8, and 10, converting every number in the training corpus to the model's base (Appendix C.1). In base 7, which is prime, CRT favors only period 7, while the harmonic account also predicts 7/2 and 7/3. Each model learns the harmonic family of its base.

A second challenge is that steering alone does not establish that a feature encodes its period. Patching a fitted plane can move the output toward a target even when the model does not use that plane (Makelov et al., 2024), and no benchmark tells whether a patching result reflects the model's own mechanism. We therefore fix the test in advance and validate each native harmonic in two ways (Figures 3 and 4). First, we compare against control families of non-native periods, fitted and patched identically. The controls do not decode their own periods, steer much less than the native harmonics, and steer in proportion to their overlap with the native subspace. Second, we examine not just whether a harmonic steers toward the target digit, but what it writes into all digit scores. Each native harmonic produces a periodic pattern over the digit logits that matches its own residue classes. Period 5 raises the two digits that are equal mod 5 and period 2 raises digits of the target's parity, while no control writes a pattern that follows its own period. Each harmonic carries only its own stops, and the digit is fixed where the stops of the whole family meet.

We then apply what we learn to Qwen3.5-9B-Base. Interchange interventions that swap feature angles between inputs test which digits each feature favors and how their contributions combine, building on work that tests the role of geometric coordinates through interventions (Engels et al., 2025; Kantamneni & Tegmark, 2025; Gurnee et al., 2025; Sarfati et al., 2026; Wurgaft et al., 2026). Near the output, these contributions to digit scores add approximately in all four trained models and in Qwen, while earlier Qwen layers reveal nonlinear interactions in which the combined effect exceeds the sum of individual effects. We also trace carry through these features and identify a small MLP population that writes both units-place and tens-place clocks jointly. Replacing these neurons' activations transfers carry-dependent predictions between inputs, showing that interlocking clocks can implement carry without a separate carry bit.

2 Setup and Method

Models. We pretrain matched 126M-parameter NanoChat models in bases 6, 7, 8, and 10. The models share an architecture, a training recipe, and a corpus in which every number is rewritten in the model's base (Appendix C.1). We compare the base-10 model with Qwen3.5-9B-Base (Qwen Team, 2026) and replicate selected results in Pythia-6.9B (Biderman et al., 2023).

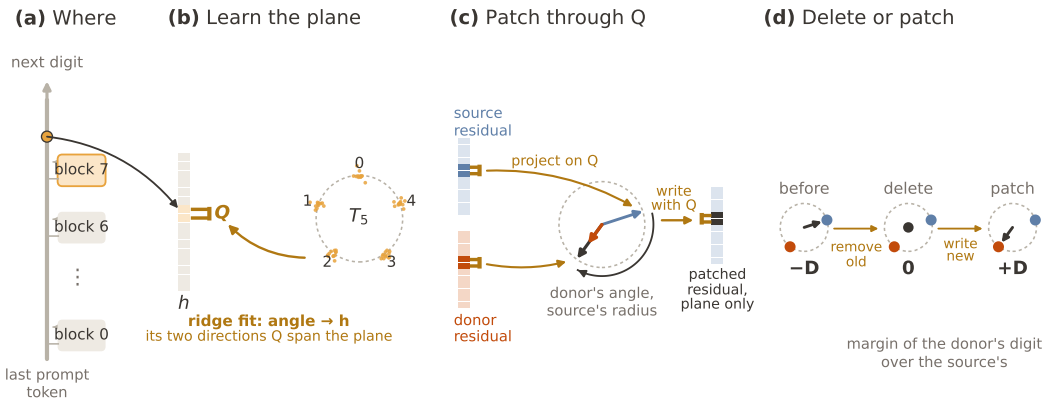


Figure 2: Method overview. (a) We read and patch the residual stream of the last prompt token after a given block. (b) For each period T, a ridge fit from the angle $2\pi n/T$ of the sum n to the activation gives the two directions Q along which the activation best tracks the angle, shown for $T = 5$. The residual is drawn as a few cells, two of which stand for Q. (c) Fourier patching projects the source and donor residuals onto Q, replaces the source's angle with the donor's while keeping the source's radius, and writes the result back through Q, leaving all other directions unchanged. (d) With the hand at the source, the margin of the donor's digit over the source's is $-D$. Deleting the harmonic removes the source's point and gives 0. Patching also writes the donor's angle, giving $+D$, so it changes the margin about twice as much as deletion.

Choice of bases. The four bases separate the two accounts to different degrees. In base 6 the accounts coincide, since the harmonic family 6, 3, 2 consists of the periods that divide the base. Base 10 adds harmonics 10/3 and 5/2, which do not divide the base. Base 8 is a power of 2, so it has no coprime factors for CRT to combine, while the harmonic account predicts 8, 4, 8/3, and 2. Base 7 is prime, so CRT favors only period 7, while the harmonic account also predicts 7/2 and 7/3. All models use a digit-level tokenizer, so each digit is a separate token and can be studied with the correct preceding digits supplied.

Fourier features and harmonic families. We study one answer digit at a time, with the correct higher digits of the answer already in the prompt. A Fourier feature with period T places the sum n on a circle:

$$\phi_T(n) = \left(\cos\frac{2\pi n}{T}, \sin\frac{2\pi n}{T}\right)^\top. \tag{1}$$

The angle on this circle is the feature's *phase*. Two sums that differ by T land at the same angle, and sums closer than T apart land at different angles.

For the units digit in base b, we test the *harmonic family* with periods b/k for $k = 1, \ldots, \lfloor b/2 \rfloor$. Harmonic k moves k clock positions each time the digit increases by one, and the *fundamental* ($k = 1$) has period b. In base 10 the units family has periods 10, 5, 10/3, 5/2, and 2. Each higher digit position multiplies the periods by the base, so the tens family has periods 100, 50, 100/3, 25, and 20. Larger k add nothing new, because harmonic $b - k$ moves k positions backwards and, at units, spans the same plane (Figure 1, right).

Finding and patching the features. For each period T, a ridge regression from the Fourier coordinates of the sum to the activation gives the two directions Q along which the activation best tracks the angle $2\pi n/T$ (Figure 2b). We fit a family from the shortest period to the longest and remove from each new plane the directions of the planes before it, so the planes of a family are mutually orthogonal. Without this step, planes of different periods can share directions, and patching one period would also turn the others, so its own contribution could not be isolated. *Fourier patching* works entirely within this plane. It projects the source and donor activations onto Q, replaces the source's angle with the donor's while keeping the source's radius, and writes the result back, leaving every other direction unchanged (Figure 2c). Any effect on the output therefore comes from these

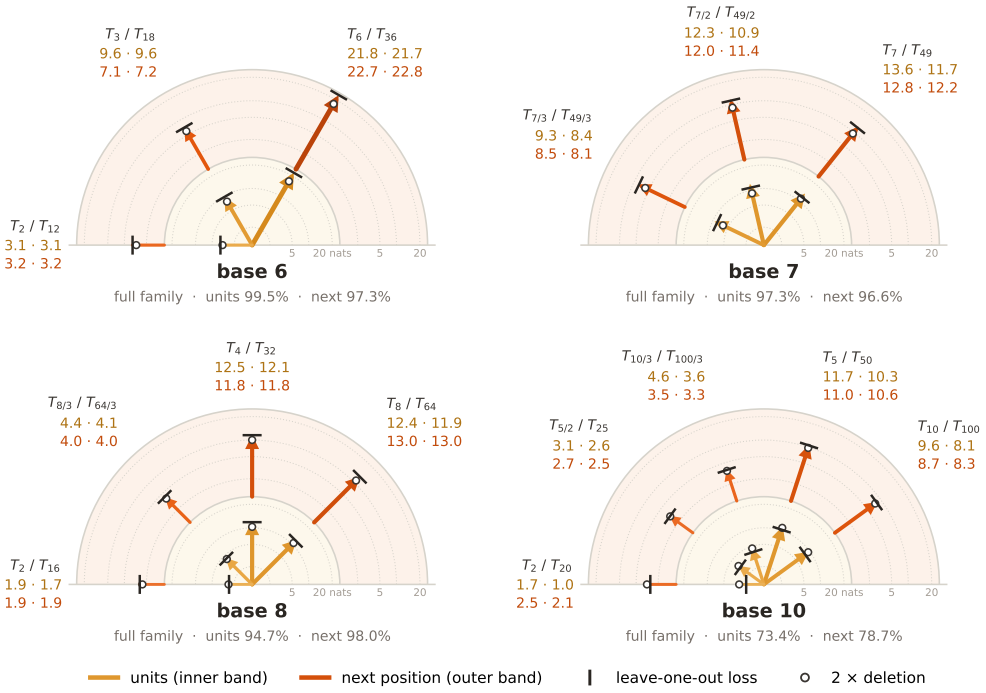


Figure 3: Each native harmonic after block 7 in the models trained in bases 6, 7, 8, and 10. Each spoke is one harmonic k, drawn at k/b of a turn. The inner band is the units digit and the outer band the next digit. The arrow gives the standalone gain (length proportional to $\sqrt{\text{nats}}$), the tick the leave-one-out loss, and the open circle twice the margin lost when the harmonic is deleted, the reference for its gain (Appendix C.4). If the harmonics interacted, the ticks would lie beyond the arrowheads instead of on them. Labels give the periods and the gain and loss in nats (units, then next); period-2 features are one-dimensional and drawn without arrowheads. Full-family target following appears under each base.

two directions, and whether they carry the period is what the controls test. Fitting and patching details are in Appendix A.1.

3 Only the Harmonics of the Base Steer Meaningfully

We patch each base's units and next-position families after block 7 in the four NanoChat models, on 1,500 initially correct cases per base and position (Appendix C.2).

Evaluation. We ask two questions of each harmonic, and for both we compare native harmonics with non-native control periods fitted and patched in the same way. **(1) How strongly does it steer?** We measure the *gain*, the change in the logit difference between the donor's and the original digit, in nats. For each harmonic, the *standalone gain* is the gain from patching it alone, and the *leave-one-out loss* is the gain lost when it is left out of the full family. Deleting a harmonic, which moves its hand to the center of the clock, measures how much the model's own computation relies on it. Patching instead turns the hand from the source to the donor, removing the source's support and giving the same support to the donor's digit, so the gain should approach twice the change deletion makes to the same logit difference (Figure 2d). This $2\times$ ratio serves as a reference for each harmonic's gain (Appendix C.4). **(2) What does it write?** A patch changes the scores of all digits, and a harmonic should raise exactly the digits at the target's stop, the target and the digit five away for period 5 and every digit of the target's parity for period 2. Section 3.1 answers (1) and Section 3.2 answers (2).

3.1 Native Harmonics Steer the Prediction, and Non-Native Periods Barely Do

The full native family changes the prediction to the donor's digit in 73.4–99.5% of cases across bases and positions, which we call *target following* (Figure 3). The contributions add approximately at the final block, where the full-family gain is within 5.0% of the sum of the standalone gains in every setting. Patching gains 0.89–1.00 times the deletion reference for full families and 0.89–1.45 times for single harmonics (Figure 3, open circles). Each family moves only its own digit position. The base-7 next-position family $T_{49}, T_{49/2}, T_{49/3}$ never changes the units prediction (0 of 1,500 cases; Appendix C.5).

Non-native control periods steer far less. Native families reach 28.6–34.7 nats of margin gain, while non-native control families reach 1.9–11.7 nats and do not decode their own periods ($R^2 \leq 0.015$ for a ridge decoder from the activation to the angle). The controls' steering tracks their overlap with the native subspace ($r = 0.80$), indicating that it comes from native directions rather than from the control periods (Appendix C.3). Base 7 provides a sharper test. Parity is well defined in every base and follows from the parities of the operands, yet base 7 has no period-2 harmonic. A period-2 plane orthogonal to the native family changes no layer-7 units prediction (0 of 1,500 cases). The model uses the harmonics of its base, not every periodic feature the task defines.

3.2 Native Harmonics Write Their Own Stops, and Non-Native Periods Do Not

Figure 4 shows what each patch writes into the units-digit scores, for native harmonics and non-native control periods on a common scale. Each native harmonic places its maxima on the digits at the target's stop. Period 5 in base 10 raises t and $t + 5$, period 4 in base 8 raises t and $t + 4$, period 3 in base 6 raises t and $t + 3$, and every harmonic in base 7 peaks at the target alone. On the same scale, the non-native control periods are nearly flat and do not write patterns that follow their own periods. The next digit shows the same picture (Appendix C.3, Figure 16), and Figure 15 rescales each units row to its own peak.

4 Fourier Feature Interactions and the Carry

Leave-one-out patching asks whether harmonics act together. If they add, removing one from the family loses about what that harmonic gains alone, and its tick sits near its arrowhead. If the loss is larger, the harmonic is stronger together with the others than alone. In the trained models after block 7, every tick sits at or just inside its arrowhead (Figure 3), so there the harmonics simply add. Qwen3.5-9B-Base computes the answer over many more layers, which lets us ask whether this holds at every depth. We run the test across its layers, and then ask how the carry enters the tens-place harmonics.

4.1 Earlier Patches Reveal Nonlinear Interactions

At each block from 21 to 31 of Qwen, we fit the five harmonics of the units, tens, and hundreds digits with the protocol used for the trained models and patch each harmonic alone, every leave-one-out subset, and all five together on 800 initially correct cases per position (Appendix B.2).

After block 31 the harmonics add, as in the trained models (Figure 5, left top). Every leave-one-out tick sits on its arrowhead, the full family's gain equals the sum of the standalone gains at all three positions, and each harmonic writes the pattern of its stops (Figure 5, right). After block 28 the harmonics interact (Figure 5, left bottom). The ticks lie beyond the arrowheads, and the full family gains 1.6–2.2 times the sum of its parts. Earlier still, the units harmonics can already be decoded, yet the full family barely changes the units prediction before block 28 (Figure 9). Two further controls are in Appendix D. In Pythia-6.9B, patching the harmonics of counterfactual operands at the operand tokens moves the answer's units digit to the one they imply in 88% of cases, against 9% for shuffled operands, so the harmonics are not only an output mechanism. In Qwen, taking two harmonics' angles from two different donors steers the model to a digit that neither donor's sum contains.

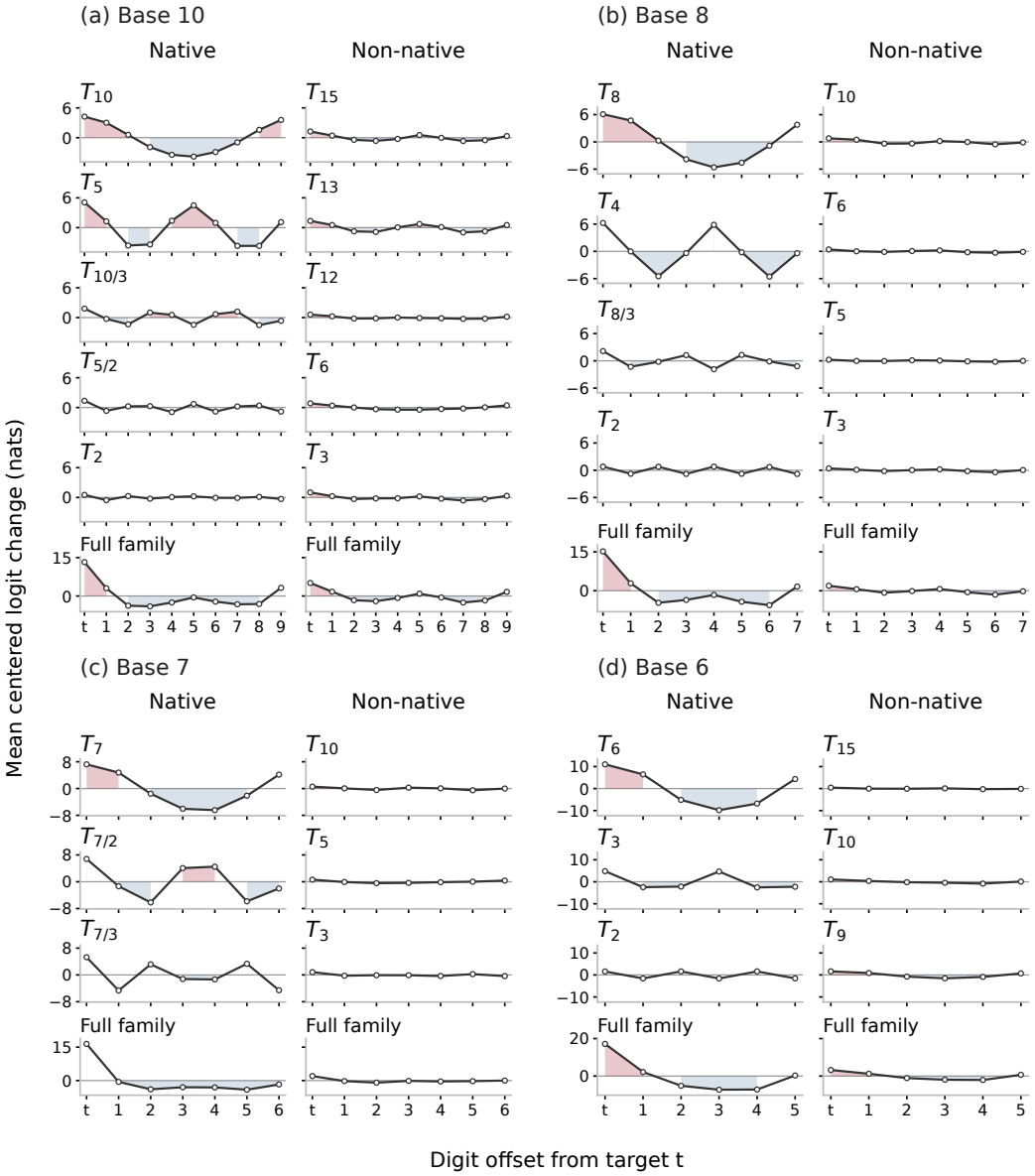


Figure 4: What native harmonics and non-native control periods write into the units-digit scores after block 7, on a common scale within each model (one scale for single periods and one for full families; $n = 1{,}500$ per model). Each row is the change in the digit logits, centered across digits, aligned to the target digit t, and averaged over cases. Native rows are the harmonic family of each base; non-native rows are the control family of Table 5.

4.2 Carry Advances the Tens Clocks

A one-hot code for each digit would need a separate carry bit that tells the tens-place computation to add one. Harmonic features offer another option. Like the hour hand of a wall clock, which moves a twelfth of a turn while the minute hand completes one, each tens-place harmonic can absorb the carry by turning as far as a one-digit increase predicts, $k/10$ of a turn for harmonic k. We test this in Qwen with two kinds of paired additions (Appendix B.5).

The first kind of pair shares the tens digits of the operands but differs in carry, as in $12 + 25 = 37$ and $18 + 29 = 47$. Both have tens sum $1 + 2$ and units digit 7, but only the second carries, so its

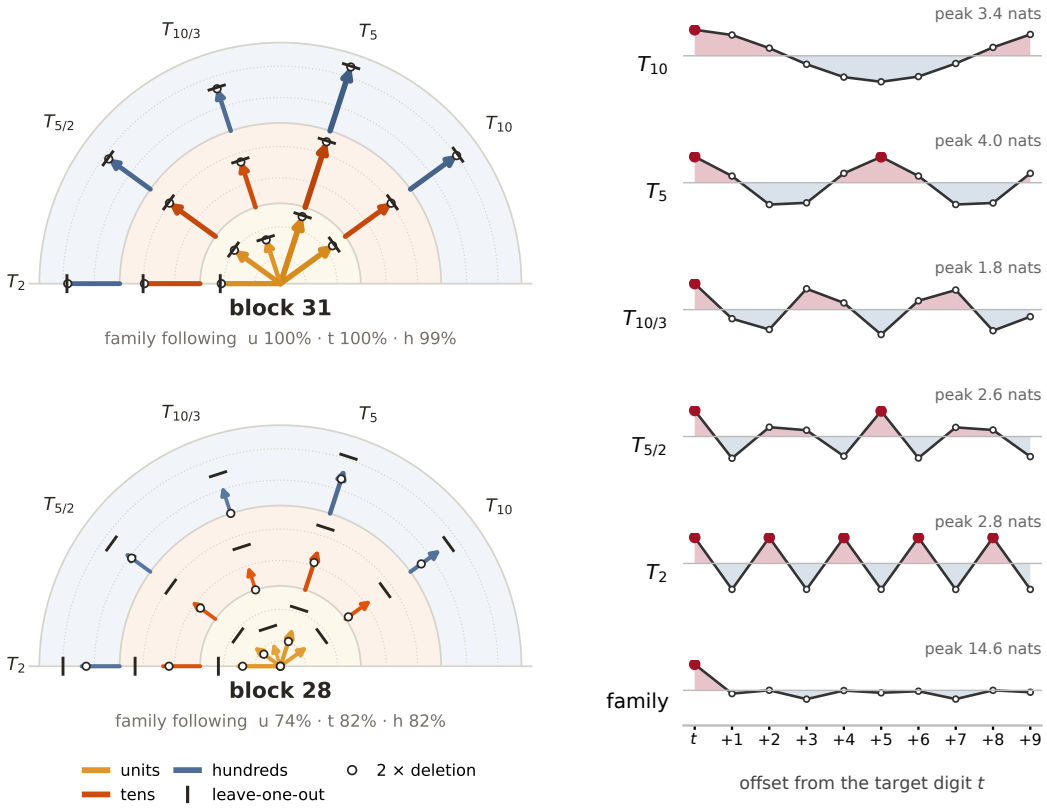


Figure 5: Figure 3 for Qwen, with three bands for the units, tens, and hundreds digits, after the additive block 31 (top) and the superadditive block 28 (bottom). At block 28 the ticks lie beyond the arrowheads, farther from the center, because leaving one harmonic out costs more than that harmonic gains alone. At block 28, deleting T_{10} or $T_{10/3}$ alone leaves the margin unchanged, so their open circles sit at the center. Periods are for units and are multiplied by 10 and 100 at tens and hundreds. Right, what each harmonic writes into the units digit scores after block 31, as in Figure 15. Tens and hundreds give the same patterns (Figure 10); numbers are in Table 1.

tens clocks should be one stop ahead. Over 900 such pairs, the mean paired differences after block 30 point along this one-stop correction in all five tens planes, within 7°, and reach 84–93% of its size (Appendix B.5, Figure 13a,b). The parity plane T_{20} moves last. It follows the tens sum without carry at blocks 23–24 and takes up the carry across blocks 25–27, while its radius briefly shrinks (Figure 13c).

The second kind of pair reaches the same answer with and without carry, as in $12 + 25 = 37$ and $18 + 19 = 37$. If the carry is fully absorbed, the tens clocks of the two inputs should agree. Over 1,000 such pairs, their phases after block 30 differ by at most 6° in four planes, while the parity plane still lags by 24° (Table 3). The tens clocks thus move one stop with the carry and meet when the answers agree. We do not test whether a separate carry signal also exists elsewhere in the model.

5 MLP Populations Write the Clock Features

The interventions above edit the residual stream. We next ask which MLP neurons write the clock features. Rotating a phase amounts to a complex multiplication, which makes the MLP's multiplicative gate a natural candidate (Nanda et al., 2023). On discovery contexts, we score each neuron by its contribution to each harmonic plane and keep a small population (Appendix B.3).

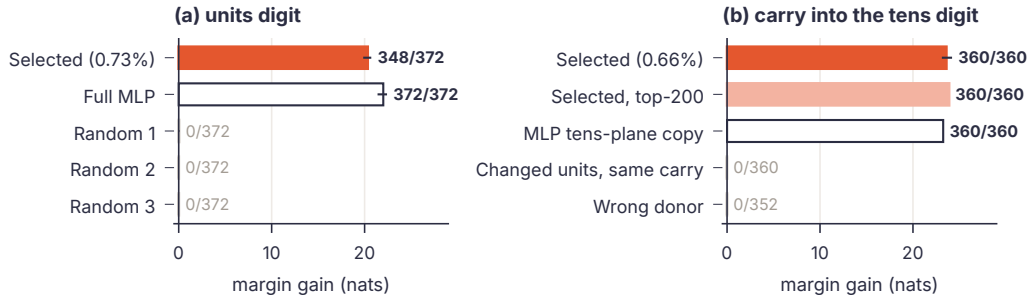


Figure 6: MLP activation replacement across L21–31. Bars show mean margin gains; labels show target-following counts for (a) units and (b) carry. Population percentages refer to neuron slots in L21–31. Shown error bars are 95% context-bootstrap intervals; missing bars do not imply zero uncertainty (Appendices B.3 and B.4).

5.1 Units-Digit Steering

The units population holds 0.73% of the MLP neurons in blocks 21–31. At the token before the units digit, we replace their gated activations, $\mathrm{SiLU}(\mathrm{gate}) \odot \mathrm{up}$, with the donor's values in every block from 21 to 31. This recovers 92.4% of the margin gain of replacing the whole MLP and moves the prediction to the donor's digit in 93.5% of held-out pairs, against 100% for the whole MLP and 0% for three random populations of the same size (Figure 6a). At block 23, a single neuron, 1588, contributes about 57% of the net parity coefficient, and its parity comes from products of the two operands, consistent with an XOR-like computation.

Figure 6a shows that this population recovers 92.4% of the full-MLP margin gain, with 93.5% target-following versus 100% for the full MLP; all three random populations yield 0% target-following.

5.2 The Tens Clock Is Analog Early and Digital Before the Readout

The output reads each clock by its stops, as a classifier over digits (Section 3.2). A tens clock that followed the full sum continuously, at one tenth of the rate of the units clock, would sit halfway between two tens stops whenever the units sum is five. We therefore hypothesized that the tens clock is analog in earlier blocks and is moved onto a stop before the readout, and that the carry appears in this step.

We test this with a separately selected population of 0.66% of the MLP neurons that writes into both the units and the tens clocks before the tens digit. Summed over blocks 21–31, its write turns the units clock T_5 by $71.8°$ and the tens clock T_{50} by $7.3°$ per unit of the operand units sum, a ratio of 9.8 against the 10 expected of meshed clocks (Appendix B.4). Split by depth, the write behaves as hypothesized (Figure 7). In blocks 21–24 the tens write follows the units write at one tenth of its rate, so the tens clock tracks the full sum continuously, like an analog hour hand. In blocks 28–31 the tens write stays in place while the units sum is below ten and then moves about one stop, like the digit wheel of an odometer, and this step is the carry. These are phases of the MLP writes, not of the residual clocks, but the residual T_{50} shows the same transition from a ramp at block 24 to a step at block 30 (Figure 12b).

Replacing the population's activations with those of a donor with the opposite carry, in blocks 21–31, changes the tens prediction to the donor's in all 360 pairs, with a mean margin gain of 23.6 nats (Figure 6b). Donors with the same carry change none (Appendix B.4). The same neurons thus write both clocks and carry the carry between inputs. This shows that meshed clocks can implement carry, not that the model has no separate carry signal.

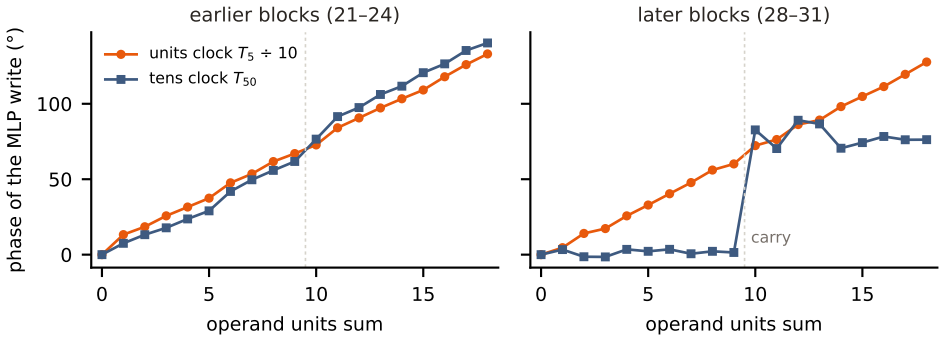


Figure 7: Phase of the selected population's MLP write at the token before the tens digit, summed over blocks 21–24 (left) and 28–31 (right) and read in the block-27 planes, as the operand units sum goes from 0 to 18 (mean over four held-out contexts, each aligned to its value at 0). The units clock T_5 is divided by 10. In earlier blocks the tens clock turns at one tenth of the rate of the units clock. In later blocks it stays in place until the units sum reaches ten and then moves about one stop.

6 Related Work

Algorithms learned from arithmetic. Studies of models trained on arithmetic have recovered Fourier algorithms for modular addition and examined how architecture and training affect which frequencies contribute (Nanda et al., 2023; Zhong et al., 2023; Ding et al., 2024; Furuta et al., 2024; McCracken et al., 2025). Work on multidigit addition and subtraction traces how circuits combine computations across digit positions (Quirke & Barez, 2024; Quirke et al., 2025).

Numerical reasoning in pretrained language models. Pretrained language models represent numbers and cyclic concepts with periodic features, including periods tied to decimal digits (Engels et al., 2025; Levy & Geva, 2025; Zhou et al., 2024; Kantamneni & Tegmark, 2025; Kadlčík et al., 2025; Fu et al., 2026; Feucht et al., 2026). Fourier features also support precise number embeddings designed for language models (Zhou et al., 2025). Related accounts trace how MLPs contribute to arithmetic predictions (Stolfo et al., 2023; Nikankin et al., 2025; Baeumel et al., 2025), with further work linking carry to continuous geometry (Wen et al., 2026).

Training data and representation geometry. Correlations and symmetries in training data shape the geometry of learned representations (Saxe et al., 2019; Korchinski et al., 2025; Karkada et al., 2026; Prieto et al., 2026). Such geometry can encode relations between concepts and support operations such as counting (Engels et al., 2025; Park et al., 2025b;a; Gurnee et al., 2025; Sarfati et al., 2026; Wurgaft et al., 2026).

Causal analysis of neural network internals. Causal mediation analysis and causal abstraction use interventions on internal states to test their role in a model's behavior (Vig et al., 2020; Finlayson et al., 2021; Meng et al., 2022; Stolfo et al., 2023; Mueller et al., 2026; Geiger et al., 2025b; 2024; Wu et al., 2023; Geiger et al., 2025a). Our phase patches build on interventions that use the geometry of a representation to control model predictions (Engels et al., 2025; Kantamneni & Tegmark, 2025; Gurnee et al., 2025; Sarfati et al., 2026; Wurgaft et al., 2026). Algebraic analysis and equivariance also help recover explicit algorithms from trained networks (Lutz et al., 2025; 2026).

7 Discussion

Numeral notation organizes arithmetic features into harmonic families. Their approximately additive contributions near the output coexist with nonlinear interactions earlier in Qwen. Linear superposition of geometric features (Klindt et al., 2026; Engels et al., 2025) therefore need not imply linear computation. In our setting, parity composition is addition of phases but multiplication of the operand phasors, $(-1)^a(-1)^b = (-1)^{a+b}$, consistent with the MLP's cross-operand products.

Differences from small modular-addition models (McCracken et al., 2025) also motivate testing proposed mechanisms across training settings.

Joint MLP writes across decimal places likewise support a clock-based account of carry, without excluding a separate carry computation.

Decodability alone does not identify causal control. In Qwen, the units harmonics decode their periods from block 21 onward, yet the full family barely changes the units prediction before block 28 (Section 4.1). We therefore suggest treating the harmonic family, rather than one prominent plane, as the unit of analysis. Recent work asks whether sparse autoencoders recover concept manifolds rather than isolated directions (Bhalla et al., 2026). Our radix models offer predicted harmonic periods and intervention-tested features for evaluating whether a dictionary recovers the full family.

Limitations. We identify causal features and contributing MLP populations, not a complete operand-to-answer circuit. Phase patches use natural counterfactual directions and do not map intermediate-angle decision boundaries. The Qwen layer sweep covers units, and its harmonics coprime to ten are spectrally less pure. The radix models are small, use one seed per base, and are trained with explicit arithmetic. Primary interventions are near the output, and cross-position and ceiling controls use earlier uncentered runs in bases 7 and 8 only. Target-following is a coarse endpoint, and several intervals are exploratory and unadjusted for multiple comparisons.

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AI USE STATEMENT

Large language models were used most heavily for coding, figure generation, and writing, and we describe our methodology for each. For code, the authors designed the experiments and wrote the core intervention logic; language models were then used as coding agents to implement analysis pipelines, batch runners, figure scripts, and validation scripts around that core. The validation scripts test what the authors specified: intervention geometry and the intervention math on synthetic inputs, targeting of the intended layer and token position, preservation of the same evaluation examples across layers, correct resumption of interrupted runs, and consistency checks that recompute the reported statistics from saved outputs. A run was accepted only when these tests passed. For analysis, the experiments produced a large archive of logged outputs; language models were used to search this archive for patterns and to propose additional analyses of the saved data, which the authors decided whether to run and then reviewed. For writing, language models drafted and polished sections of the manuscript under the authors' direction and were used to check reported numbers against saved outputs. The research questions, experimental design, models, datasets, interpretation, and claims are the authors' own, and the authors take full responsibility for the work and its final content.

REPRODUCIBILITY STATEMENT

The appendix reports the radix-training corpus, model and training configuration, qualification criteria, data splits, and Fourier analysis procedure. The main text specifies the intervention, controls, evaluation sets, and uncertainty estimates used for each causal result.

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A HARMONIC FAMILIES AND FITTING

A.1 PROBE FITTING, PATCHING, AND SCORING

Probes.

- Let $h \in \mathbb{R}^D$ be the residual activation after a chosen block at the token immediately before the digit being predicted, with D the residual width. For period T, $\phi_T(n) \in \mathbb{R}^r$ with $r = 1$ for $T = 2$ and $r = 2$ otherwise.
- Two affine ridge maps decode Fourier coordinates and estimate activation directions for patching:

$$\phi_T(n) \approx \mathbf{B}h + a, \qquad h \approx \mathbf{W}\phi_T(n) + \mu, \qquad \mathbf{Q} = \mathrm{orth}[(\mathbf{I}_D - \mathbf{P})\mathbf{W}], \tag{2}$$

  with $\mathbf{B} \in \mathbb{R}^{r\times D}$, $a \in \mathbb{R}^r$, $\mathbf{W} \in \mathbb{R}^{D\times r}$, and $\mu \in \mathbb{R}^D$.
- Each fit minimizes squared error plus the squared Frobenius norm of the coefficient matrix, with penalty $\alpha = 1$ and unpenalized intercepts. Decoding performance is the mean held-out R^2 over the predicted coordinates.
- $\mathbf{P}$ is the orthogonal projector onto protected directions ($\mathbf{P} = \mathbf{0}$ when none are protected). Within a family, the protected directions are the planes of all shorter periods, plus the lower-position fundamental T_b at the next digit. orth returns an orthonormal basis $\mathbf{Q} \in \mathbb{R}^{D\times q}$, where $q \le r$ is the retained numerical rank.
- After the protected directions are removed (the projection is applied twice for numerical stability), orth keeps the left singular vectors of the residualized coefficient matrix whose singular values exceed $\max(10^{-10},\, 10^{-6}\,\sigma_{\max})$, where $\sigma_{\max}$ is the largest singular value of the raw matrix.
- Patching subspaces are fitted on sums disjoint from the evaluation cases. Period grids, splits, fitting orders, and protected directions are specified with each experiment.

Fourier patching.

- Paired inputs match on the answer prefix, the lower digits, and the incoming carry, and differ only in the digit being tested.
- For original and donor activations h_o, h_c and subspace center $c \in \mathbb{R}^q$, let $z_o = \mathbf{Q}^\top h_o - c$ and $z_c = \mathbf{Q}^\top h_c - c$. The patched activation is

$$\widetilde{h}_o = h_o + \mathbf{Q}\left(\|z_o\|_2 \frac{z_c}{\|z_c\|_2} - z_o\right), \tag{3}$$

  which replaces the angle, keeps the radius, and leaves the orthogonal complement of the plane unchanged. In a one-dimensional subspace (period 2) it replaces the sign and keeps the magnitude.
- Centered patches use $c = \mathbf{Q}^\top\mu$, with μ the reverse-ridge intercept from the fitting rows; uncentered patches use $c = 0$. Each experiment states which it uses.
- For a harmonic subset S, the displacements of the individual planes, each computed from the unedited activation, are summed; $S = \varnothing$ is the baseline.
- Fourier patching adds no magnitude: unlike steering with a chosen gain (for example, a gain of ten in Feucht et al. (2026)), it is an interchange intervention on the angle at the activation's own radius.

Scoring.

- Target following uses the highest-scoring token over the full vocabulary.
- With original digit s, target digit t, and $\ell_S(d)$ the logit of digit d after patching the harmonics in S, the logit difference score and its gain are

$$m(S) = \ell_S(t) - \ell_S(s), \qquad \Delta m(S) = m(S) - m(\varnothing). \tag{4}$$

  For a family F, the standalone gain of harmonic k is $\Delta m(\{k\})$ and its leave-one-out loss is $\Delta m(F) - \Delta m(F \setminus \{k\})$.

- Joint patches are compared with the additive prediction $\sum_{k \in S} \Delta m(\{k\})$, with s and t fixed across subsets within each case. When the supplied phases identify several digits, predictions are examined over that set.
- These definitions apply to the NanoChat and Qwen experiments, which supply the answer prefix and score single-digit tokens. Pythia predicts a whole one-token answer (Appendix D.1).
- Interventions act after the final NanoChat block, indexed as block 7, and after block 30 in Qwen, chosen by a held-out layer sweep, unless stated otherwise.

A.2 Phase Separation and Information Content of Harmonics

Write the sum as $n = bpq + pd + r$, where $p = b^j$ is the place value of the digit d at position j ($j = 0$ at units), q collects the higher digits, and $0 \le r < p$ is the lower suffix. The family at position j has periods bp/k. Changing only the digit d changes the phase of the harmonic with period bp/k by

$$\Delta\theta_{bp/k} = \frac{2\pi k\, p\, \Delta d}{bp} = \frac{2\pi k\, \Delta d}{b} \pmod{2\pi}, \tag{5}$$

which does not depend on p: the ideal separation between adjacent digit candidates is $2\pi k/b$ at every position, and the lower suffix r adds the common offset $2\pi kr/(bp)$ to all candidates. The phase of harmonic k is $2\pi kd/b$ up to this offset, so two digits d, d' give the same phase exactly when $k(d - d') \equiv 0 \pmod{b}$, that is, when $d \equiv d' \pmod{b/\gcd(k, b)}$. A single harmonic therefore determines d modulo $b/\gcd(k, b)$, a set of harmonics determines it modulo the least common multiple of these moduli, and the joint ideal phase labels uniquely identify the digit exactly when that least common multiple equals b. This is an information criterion and does not by itself guarantee successful steering; the base-7 single-harmonic results are the clearest example. In base 10, $\{T_2, T_5\}$ gives $\mathrm{lcm}(2, 5) = 10$; in base 8, $\{T_2, T_4\}$ gives $\mathrm{lcm}(2, 4) = 4 < 8$, leaving the pair $\{t, t + 4\}$ unresolved; in base 7, every harmonic alone gives modulus 7.

B Qwen Decimal Experiments

All Qwen experiments use Qwen3.5-9B-Base on six-digit addition with the answer written digit by digit. Unless stated otherwise, interventions act on the residual stream after block 30 at the token immediately before the digit being predicted, subspaces are fitted on 4,000 baseline-correct examples and evaluated on 800 baseline-correct examples from disjoint sums. Intervals are paired and unadjusted for multiplicity; the layer sweep uses 2,000-draw paired bootstraps, and the carry analysis uses cluster bootstraps over pair groups.

B.1 Held-Out Fourier Probes

The probes reverse the direction of the fit used for interventions, mapping activations to Fourier labels, and measure linear decodability rather than causal use. For every period $T \in \{2, \ldots, 1000\}$ we fit affine ridge probes ($\alpha = 1$) from the residual state after each block to $\sin(2\pi n/T)$ and $\cos(2\pi n/T)$, with n the full sum, at the positions before the hundreds, tens, and units digits. A fixed split (seed 42) holds out 20% of each magnitude grid, giving 3,200 fitting and 800 evaluation examples. We report the mean held-out R^2 of the two targets (cosine only for $T = 2$); negative values are kept in the results and clipped to zero only in the heatmaps.

The strongest readability at the hundreds position is selective. Of the 999 periods scanned, only twelve reach held-out $R^2 \ge 0.95$ at any layer, namely $200, 250, 364, 400, 445, 500, 576, 809, 973, 978, 989$, and 1000, all peaking at layer 31; the nominated hundreds family $T_{200}, T_{500}, T_{1000}$ is among them ($R^2 = 0.954, 0.983, 0.991$). The threshold is descriptive, and because fitting and evaluation examples share magnitude grids the R^2 values measure within-grid generalization.

B.2 Full Harmonic Factorial

Orthogonal layer sweep. The sweep in Section 4.1 patches the last-token block output separately at each block from 21 to 31, for units, tens, and hundreds. Periods are 2, $5/2$, $10/3$, 5, and 10

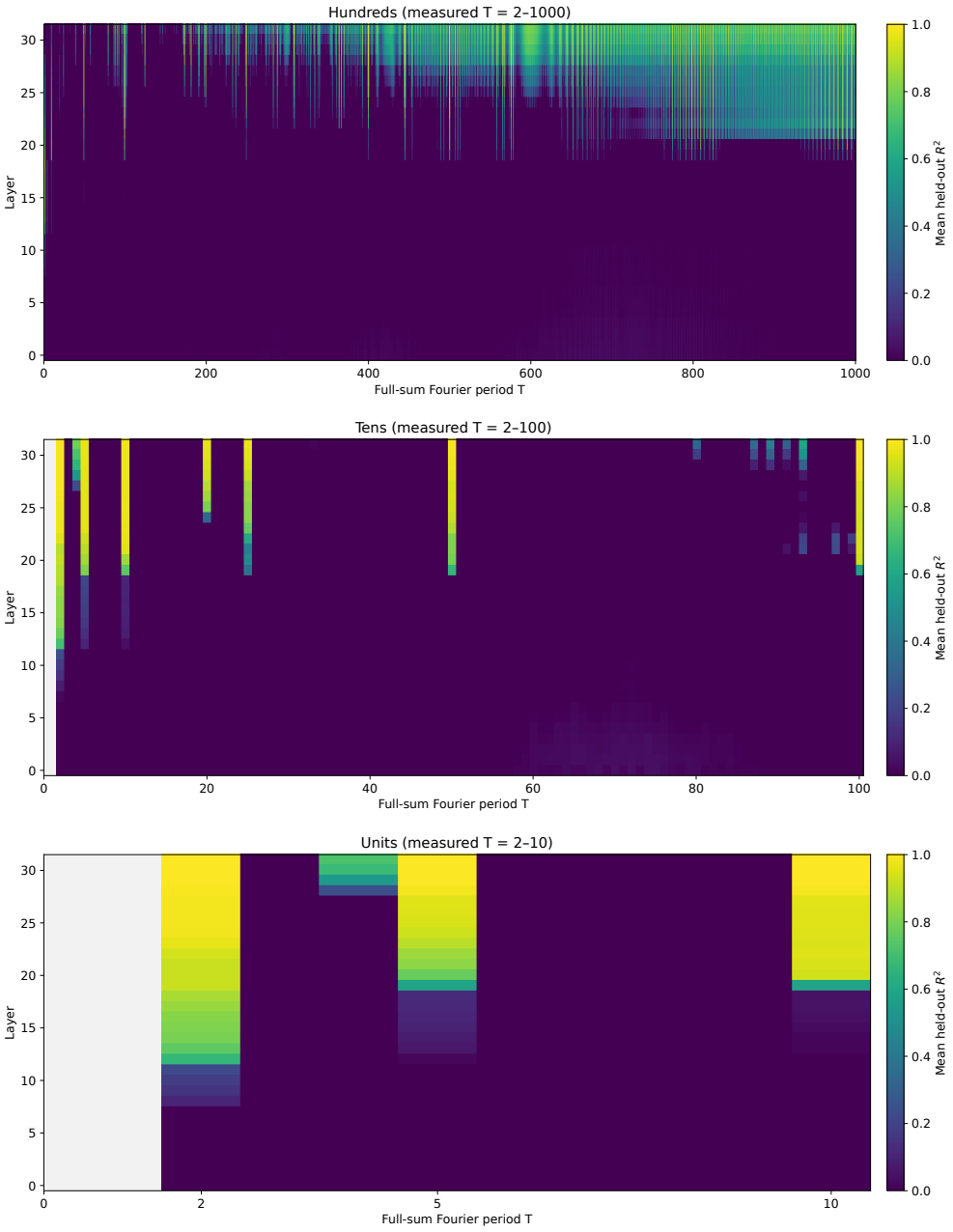


Figure 8: Held-out Fourier-probe R^2 before the hundreds, tens, and units digits (top to bottom), for periods 2–1000, 2–100, and 2–10 across layers 0–31. Periods are not rescaled by position; the colour scale is clipped to $[0, 1]$.

times the place value. Planes come from a centered label-to-activation ridge fit ($\alpha = 1$, unpenalized intercept) and are orthogonalized from the shortest period up, protecting T_{10} at tens and T_{100} at hundreds, as in the trained models. The edits replace each plane's angle with the donor's and keep the source radius. The arms are the baseline, an exact self-patch, five singles, five leave-one-out subsets, the full family, and the sum of raw-plane edits, which gives the same full family gain as the orthogonal family to within 1 nat at every block and position. Fit and evaluation prompts are the canonical 4,000 and 4,000 of the Qwen grid. The 800 source cases per position are balanced over four operand blocks, ten source digits, and shifts of 1, 3, 7, and 9, and match the prefix, lower suffix, and incoming carry. At every block and position, the self-patch reproduces the full-vocabulary logits exactly. Decoding R^2 comes from a separate activation-to-label probe on all 4,000 evaluation prompts. Intervals are paired case bootstraps. The excess of the full family over the summed singles excludes zero at every block from 21 to 30 and is within 0.05 nats of zero at block 31.

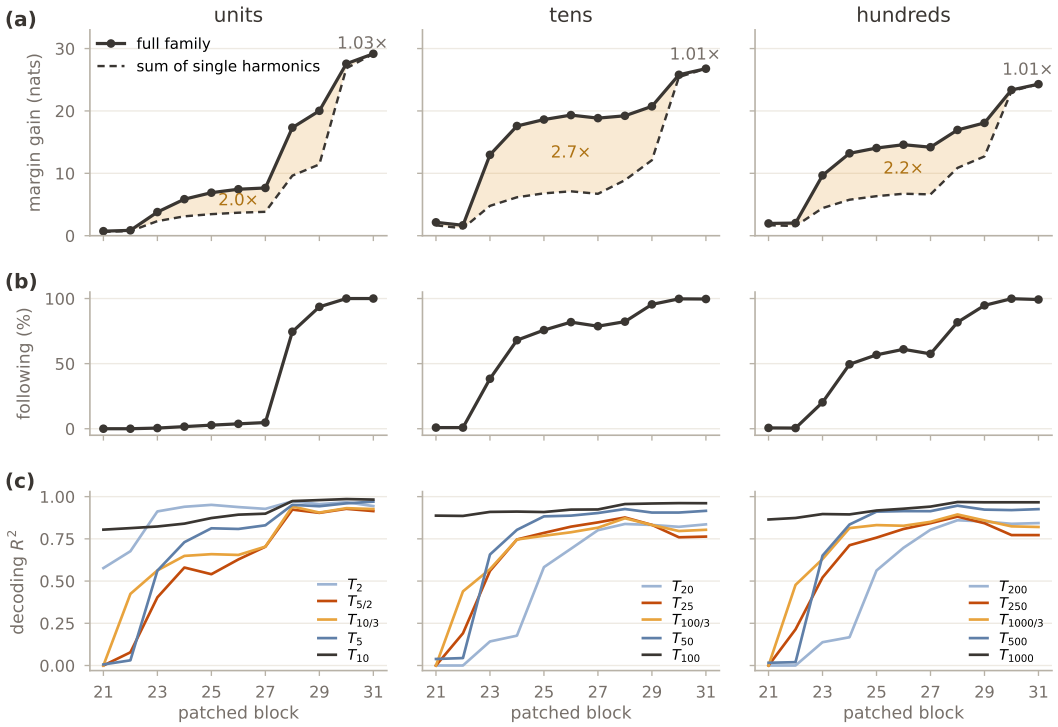


Figure 9: Harmonic interactions across Qwen layers ($n = 800$ cases per position). (a) Margin gain of the full family and the sum of the five single-harmonic gains; shading marks the excess, with the ratio at block 26 and block 30. (b) Target following of the full family. (c) Held-out decoding R^2 of each harmonic plane. Each block uses planes fitted at that block.

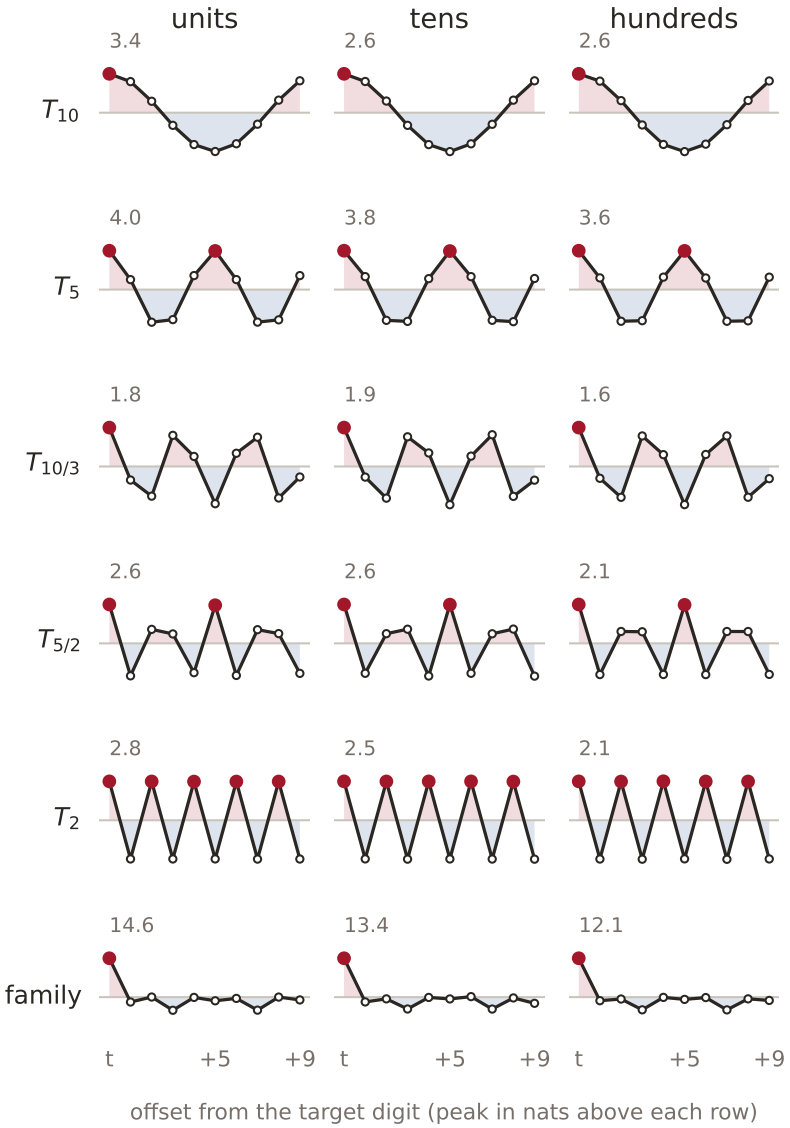


Figure 10: What each harmonic writes into the digit scores after block 31 in Qwen, at units, tens, and hundreds ($n = 800$ per position). Rows are scaled by their value at the target digit, printed above each row in nats; red points mark digits on the target's stop. Periods are for units and are multiplied by 10 and 100 at tens and hundreds.

Table 1: Qwen harmonic contributions behind Figure 5. Each cell gives the standalone gain, the leave-one-out loss, and twice the deletion change, in nats ($n = 800$ per position). Periods are for units and are multiplied by 10 at tens and by 100 at hundreds.

block	position	T_{10}	T_5	$T_{10/3}$	$T_{5/2}$	T_2	family
31	units	6.8 / 6.9 / 6.4	8.0 / 8.0 / 7.7	3.5 / 3.5 / 3.3	5.2 / 5.2 / 5.0	5.6 / 5.6 / 5.4	29.2 / – / 29.2
31	tens	5.2 / 5.2 / 5.0	7.6 / 7.6 / 7.3	3.8 / 3.8 / 3.5	5.2 / 5.2 / 5.0	5.0 / 5.0 / 4.8	26.8 / – / 26.7
31	hundreds	5.3 / 5.3 / 5.1	7.2 / 7.2 / 7.0	3.2 / 3.2 / 3.0	4.3 / 4.3 / 4.1	4.3 / 4.3 / 4.2	24.3 / – / 24.2
28	units	1.7 / 4.1 / -0.2	2.5 / 5.6 / 1.0	0.9 / 2.5 / -0.2	1.7 / 4.3 / 0.7	2.7 / 6.0 / 2.2	17.3 / – / 9.0
28	tens	1.6 / 3.7 / 0.0	2.8 / 6.6 / 1.3	0.9 / 3.1 / -0.2	1.4 / 4.7 / 0.5	2.1 / 6.5 / 1.5	19.2 / – / 10.5
28	hundreds	2.2 / 3.4 / 0.3	3.5 / 5.5 / 2.0	1.3 / 2.6 / -0.0	1.7 / 3.6 / 0.8	2.2 / 5.0 / 1.7	17.0 / – / 10.5

Population	Target-following (%)	Margin gain (nats)
Selected	93.55 [86.56, 98.39]	20.333 [19.850, 20.719]
Full MLP	100.00 [100.00, 100.00]	22.009 [21.392, 22.413]
Random 1	0.00 [0.00, 0.00]	0.050 [0.013, 0.082]
Random 2	0.00 [0.00, 0.00]	0.014 [0.003, 0.026]
Random 3	0.00 [0.00, 0.00]	0.019 [0.012, 0.025]

Table 2: MLP population results with 95% context-bootstrap intervals.

B.3 MLP POPULATION SELECTION AND REPLACEMENT

We enumerate the two operand units digits on a 10×10 grid in each of eight fixed six-digit operand contexts. Contexts 0–3 are used for discovery and 4–7 for evaluation. The correct answer prefix is supplied before the units digit. For each neuron, we compute the two-dimensional Fourier coefficient at the sum frequency (k, k) and multiply it by the neuron's down-projection writer projected onto the corresponding harmonic plane. We rank neurons by the norm of this complex contribution averaged over discovery contexts. For each plane, we retain neurons until reaching 90% signed coverage of its net coefficient, capped at 50 neurons, then take the union across the five planes. The resulting population contains 985 layer-neuron entries out of $11 \times 12{,}288$ across L21–31 (0.73%).

We replace selected entries of $\mathrm{SiLU}(\mathrm{gate}) \odot \mathrm{up}$ at the last prompt token with the same donor's natural values at each patched layer. The 372 held-out pairs retain context, incoming carry and answer prefix while changing the units digit by $+7 \pmod{10}$. Full-MLP replacement uses all 12,288 entries at each layer. Three fixed random populations match the selected count per layer and exclude the selected neurons. Confidence intervals use 2,000 context-bootstrap draws over the four held-out contexts. Exact self-patches reproduce logits on the eight validation examples checked for each arm.

The paired selected-minus-full margin difference is -1.676 nats $[-1.797, -1.555]$. The 92.4% ratio compares mean margin gains and is not a fraction of the model's computation. For L23 neuron 1588, T2 accounts for 99.818% of sum-diagonal power. Writing $G = \mathrm{DFT}(\mathrm{SiLU}(\mathrm{gate}))$ and $U = \mathrm{DFT}(\mathrm{up})$, the cross-operand terms $G[5,0]U[0,5] + G[0,5]U[5,0]$ account for 102.95% of its held-out parity coefficient, with opposing terms reducing the net coefficient. The parity coefficient has consistent phase across all eight contexts. This supports an XOR-like parity computation, rather than an exact Boolean gate characterization of every activation.

B.4 CARRY POPULATION AND WRITE-PHASE MEASUREMENTS

Neuron replacement. The before-tens experiment uses a separately selected population, not the units-steering population. Top-50 lists for each tens harmonic are unioned within each layer, giving 897 layer-neuron entries across L21–31. The grid varies both operand units digits from 0 to 9 in eight fixed upper-operand contexts. Discovery uses contexts 0–3 and evaluation uses contexts 4–7. The correct answer prefix is supplied and is constant within each context. The 360 primary pairs have equal units remainders and opposite carry states, with 180 pairs in each direction. Natural donor gated activations replace the live values at the last prompt token ($\alpha = 1$).

The all-layer top-50 union gives a mean margin gain of 23.597 nats with a 95% context-bootstrap interval of [23.107, 24.207]. The top-200 union and full-FFN tens-plane copy also follow the counterfactual target on 360/360 pairs, with mean gains of 23.893 and 23.251 nats. The changed-units/same-carry control follows the countercarry target on 0/360 pairs (mean gain -0.033 nats); the wrong-donor control follows it on 0/352 retained pairs (mean gain -0.016 nats). These are distinct controls, not additional replications of the primary comparison.

Summed writes. For Figure 7, we sum the selected top-50 population's down-projection writes over a range of blocks in residual coordinates, which all blocks share, and decode the sum with the block-27 planes, without centering. Slopes use the radius-weighted circular fit below, per held-out context. Summed over blocks 21–31, the T_5 slope is 71.8° (SD 0.2° across contexts, resultant 0.98) and the T_{50} slope 7.3° (SD 0.7°, resultant 0.97), a ratio of 9.8. Decoding with the block-31 planes gives 9.8, and summing over blocks 23–29 gives 9.2–9.7. Split by depth, the T_{50} write rises from 0

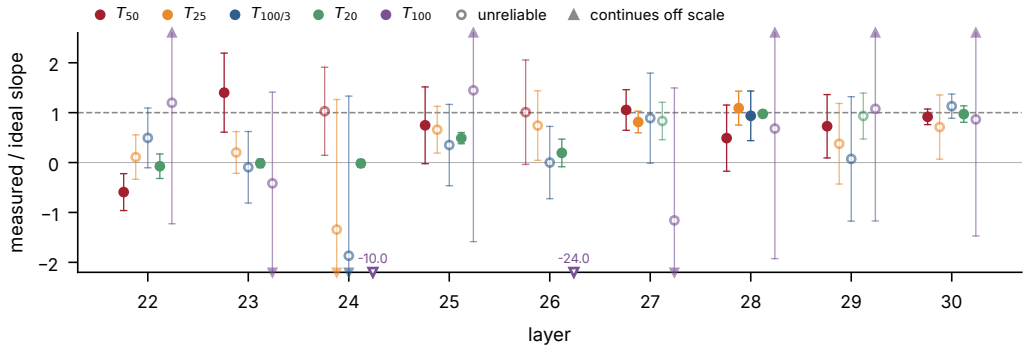


Figure 11: Selected-population tens-plane write slopes divided by ideal rates $360/T$. Bars show SD across four held-out contexts; open markers fail the reliability criterion. Triangles indicate off-scale values or intervals. The dashed line marks the ideal ratio of one.

to 62° over units sums 0–9 in blocks 21–24, against 67° for $T_5/10$, and stays within 4° of its start over 0–9 in blocks 28–31 before moving to 70–89° from units sum 10 on.

MLP write phases. Write phases are measured by applying the fit-only linear decoders to FFN increments without a centering offset. The phase slope is selected by a radius-weighted circular fit across units sums. L27 selected-population slopes are 71.59° per unit for T5 and 7.59° for T50, with mean resultants 0.86 and 0.97. Their ratio of mean slopes is 9.43; it is not a fixed ratio in every context. Both pass the recorded reliability criterion (mean resultant at least 0.8 and slope standard deviation at most 6°). This criterion measures fit consistency rather than agreement with the theoretical slope. Figure 11 shows all five tens-plane slopes across L22–30 for this population. Matching phase rates do not establish a direct causal connection between clock planes.

Residual phase profiles. For Figure 12b, we unwrap each held-out context's T50 residual phase over units sums 0–18 and subtract its phase at zero, then average the four trajectories with equal context weights. There is no smoothing or rescaling. The ideal references are $7.2°u$ for a ramp and $72°\mathbf{1}[u \geq 10]$ for a step, where u is the operand units sum. Mean within-carry-state increments decrease from 6.91° at L24 to 2.23° at L30, while the mean $9 \to 10$ increment rises from 12.33° to 50.27°. Thus neither layer is a perfect ramp or step. These residual measurements are distinct from FFN write-vector slopes in Figure 12c.

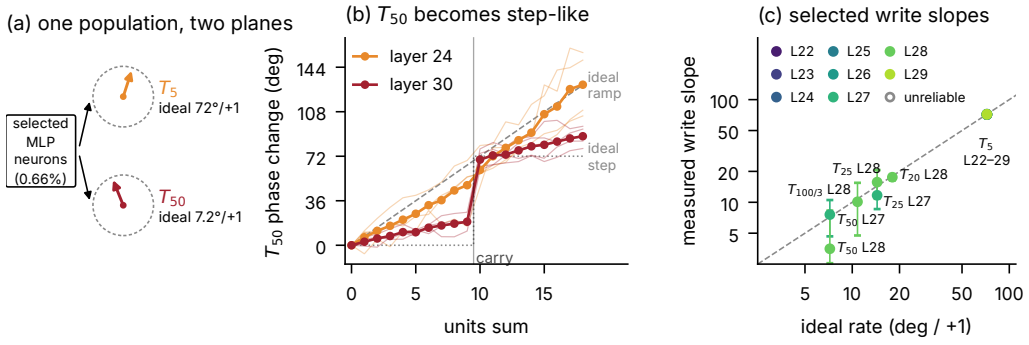


Figure 12: Shared MLP writes across decimal places. (a) Selected population and ideal phase rates; 0.66% refers to neuron slots in L21–31. (b) T50 residual phase change from units sum zero: thin lines show four held-out contexts, thick lines their mean. (c) Selected FFN write-phase slopes, with context SD bars; open markers fail the reliability criterion in Appendix B.4.

B.5 CARRY PHASE READOUT

These are the details behind Section 4.2. Phase-readout maps for each period are fitted on 1,000 carry-free examples and calibrated on a separate 1,000, and phases are read as the calibrated $\mathrm{atan2}$ of the plane coordinates, with a separately fitted basis at every layer. Pairs are matched either on the final sum (1,000 pairs) or on the local tens sum before the incoming carry (900 pairs with an unchanged answer prefix), always sharing the lower remainder; pairs at the tens-9-to-0 boundary form a separate stratum and are excluded here. Differences are carry minus no-carry, wrapped circularly, and uncertainty comes from a cluster bootstrap over the 20 pair groups. Raw planes are used throughout; the T_{20} transition is robust to using orthogonalized planes instead, whereas the onset times of some other periods are not.

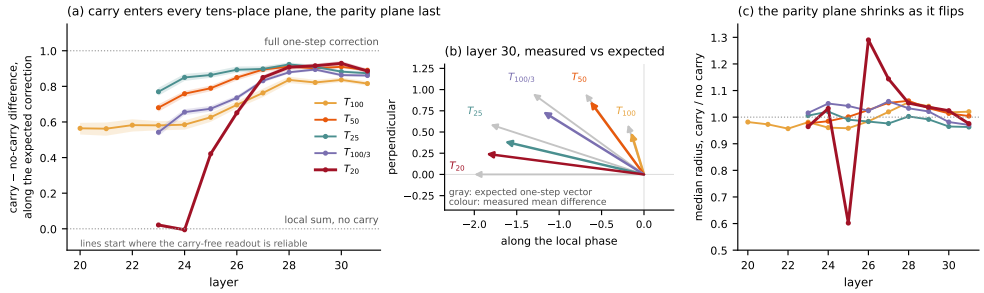


Figure 13: Carry in Qwen's tens-place Fourier features. (a) Mean carry-minus-no-carry difference projected onto the expected correction for a one-digit increase, using pairs matched on the local tens sum. A value of 1 denotes the full expected correction; curves begin where the carry-free phase readout meets the reliability criterion in Appendix B.5. (b) Measured and expected differences at block 30, expressed in each plane's coordinates. (c) Median radius on carry examples divided by the median radius on no-carry examples. Bands show cluster-bootstrap intervals over pair groups.

For the vector comparison in Figure 13, both members of a pair are rotated by their shared ideal local-sum phase, so that this phase lies on the x axis, and scaled by the no-carry median radius of that layer. In this frame the no-carry partner sits ideally at $(1, 0)$ and a one-step carry correction is the fixed vector $(\cos\delta - 1, \sin\delta)$ with $\delta = 180°, 72°, 36°, 108°, 144°$ for $T_{20}, T_{50}, T_{100}, T_{100/3}, T_{25}$. At layer 30 the angular error between measured and expected vectors is $7°$ for T_{20} and $0.4°$ to $4.7°$ for the other planes, and the coefficient along the expected vector is 0.84–0.93. These are paired representation differences, not isolated update vectors, and they do not by themselves show that all planes are corrected by one mechanism. Figure 14 gives the circular-mean phase differences for both pair designs, which agree, and the component perpendicular to the expected vector, which is small except for a transient excursion of the parity plane at layers 25–27; Table 3 lists the layer-30 values.

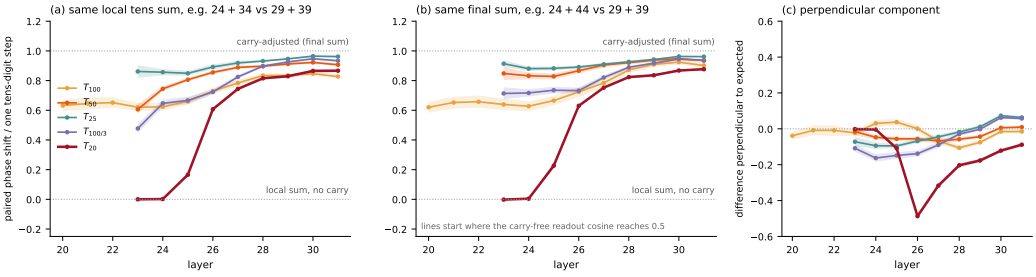


Figure 14: Carry phase differences by layer. Circular-mean paired phase difference $\Delta\theta$ in units of one tens-digit step δ: (a) $\Delta\theta/\delta$ for pairs matched on the local tens sum, (b) $1 + \Delta\theta/\delta$ for pairs matched on the final sum, on a consistent circular branch, so that the carry-adjusted prediction is 1 in both. (c) Component of the mean difference perpendicular to the expected correction. Bands are cluster-bootstrap intervals; lines start where the carry-free readout cosine reaches 0.5.

Table 3: Carry-minus-no-carry circular mean phase differences at layer 30 (degrees; raw planes).

Period	Same result	Same local sum	One-digit increment
T_{20}	-23.8	155.7	180
T_{50}	-3.8	66.4	72
T_{100}	-2.7	30.5	36
$T_{100/3}$	-6.0	102.3	108
T_{25}	-5.4	138.9	144

C Radix Experiments

The four radix models share one architecture, recipe, seed, training step (7,680), and digit-level tokenizer, and differ only in the numeral system of their training corpus. Interventions act on the residual stream at the token before the digit being predicted, with subspaces fitted on 4,000 baseline-correct rows (3,999 at sevens) and evaluated on 1,500 baseline-correct cases from disjoint sums. Primary rates are reported at the final block (L7); the native factorials are swept over L0–L7, and the earlier persistent all-layer edits are stated where used. Intervals are paired and unadjusted for multiplicity; the gate matrices use normal-approximation intervals and the two-position experiments 2,000-draw paired bootstraps.

C.1 Training Corpora and Configuration

Summary of the matched runs.

- One eight-layer NanoChat model per base $b \in \{6, 7, 8, 10\}$: decoder-only, next-token objective, residual width 512, four attention heads and four key/value heads, vocabulary of 32,768 tokens, context length 2,048.
- Shared architecture, training recipe, random seed, and digit-level tokenizer.
- 7,680 steps at sequence length 2,048 and global batch 262,144 tokens, about two billion tokens. Because numeral length varies with the base, the runs reach different points in the corpus within this budget.
- Every digit string in the equations and the web text is converted to the run's base, preserving document order and nonnumeric text, including digit strings within decimal fractions and scientific notation.

The decimal corpus is a frozen ClimbMix corpus with 9,786,141 arithmetic equations interleaved into it (Table 4). Binary addition dominates by design, and five further operation families keep the arithmetic data from reducing to a single prompt template. Equations are shuffled with a deterministic Feistel permutation (seed 42), grouped into documents of 16 equations, and inserted at approximately equal intervals into the first 150 of 170 shards (4,078 documents in shards 0–83, 4,077 in shards 84–149); shards 150–169 contain only web data.

Table 4: Arithmetic equations interleaved into the ClimbMix training corpus. Ordered pairs $a + b$ and $b + a$ are distinct records, and every binary-addition pair receives two exposures.

Family	Construction	Equations	Share
$a + b = c$	$a, b \in \{0, \ldots, 2000\}$, all ordered pairs, two exposures	8,008,002	81.83%
$a + b + c = d$	$a, b, c \in \{0, \ldots, 200\}$, sampled without replacement	1,000,000	10.22%
$a - b = c$	$0 \leq b \leq a \leq 2000$, sampled without replacement	500,000	5.11%
$a \times b = c$	$a \in \{0, \ldots, 2000\}$ and $b \in \{1, \ldots, 20\}$	40,020	0.41%
$c/b = a$	Exact inverses of the multiplication examples	40,020	0.41%
$a \bmod m = r$	$a \in \{0, \ldots, 2000\}$ and $m \in \{2, \ldots, 100\}$	198,099	2.02%
Total		9,786,141	100.00%

The base-6, base-7, and base-8 corpora are derived from the frozen decimal corpus, not from one another. Every maximal ASCII digit string is parsed as a decimal integer and rewritten in the target base in place, in the inserted equations and in numbers occurring naturally in the web text alike; shard order, document order, insertion positions, operators, and all non-numeric formatting are preserved, and digit runs inside decimal or scientific notation are converted independently. Each model is a decoder-only NanoChat transformer with eight blocks, residual width 512, four attention heads, four key/value heads, a vocabulary of 32,768 tokens, a context length of 2,048, and the `SSSL` attention-window pattern, trained with the next-token objective under the same recipe and seed in every base; the saved checkpoint configurations of the four runs agree on these settings. Each model is trained for 7,680 steps at sequence length 2,048 with a global batch of 262,144 tokens, a budget of $7{,}680 \times 262{,}144 \approx 2.0$ billion tokens. Because numeral length changes with the base, the token-level batches are not identical and the runs end at slightly different document positions.

C.2 Systematic Harmonic Patching Across Bases and Positions

For each base $b \in \{6, 7, 8, 10\}$, sums take the form $b^2q + s$, with all suffixes $s = 0, \ldots, b^2 - 1$. Fitting uses $q = 8, 10, \ldots, 36$ and evaluation uses $q = 9, 11, \ldots, 37$. For each sum, we sample 12 distinct ordered operand pairs within $[0, 2000]$, balancing carry into the next digit where feasible. Among baseline-correct examples, we select 4,000 fitting rows round-robin across sums and 1,500 evaluation pairs with unique sources, balanced across eligible block, source-digit, and target-digit combinations. Pairs match the supplied answer prefix, lower suffix, and incoming carry. The same cohort is used for independent interventions at blocks 0–7.

Reverse ridge fits use $\alpha = 1$ and an unpenalized intercept, which sets the intervention center. Periods b^{j+1}/k for digit position $j = 0, 1$ and $k = \lfloor b/2 \rfloor, \ldots, 1$ are residualized sequentially against all preceding planes; at $j = 1$, the protected span initially contains T_b. The singular-value cutoff is $\max(10^{-12}, 10^{-6}\sigma_{\max})$. If the centered source radius exceeds 10^{-10}, a donor radius at or below 10^{-10} stops the run. Fits, centers, and individual edit vectors remain fixed across subsets. The raw-family comparison uses the same ridge fits and pairs, summing component edits in the unprojected planes. Each layer includes a self-patch check of the full output logits.

Layer-7 results. The full native family reaches 99.5%, 97.3%, 94.7%, and 73.4% target following at units in bases 6, 7, 8, and 10, and 97.3%, 96.6%, 98.0%, and 78.7% at the next position, with mean margin gains of 28.6–34.7 nats. The full-family gain differs from the sum of standalone gains by at most 5.0% (base-10 units) and by at most 1.2% in five of the eight settings. In base 6 the fundamental contributes 21.8 of 34.7 nats at units; in base 7 the three harmonics give 9.3–13.6 nats; in base 8 the fundamental and period 4 give 12.4 and 12.5 nats and parity 1.9; in base 10 period 5 gives 11.7 nats, the fundamental 9.6, and parity 1.7. The same ordering holds at the next position.

Digit-score profiles. In Figure 15, period 3 in base 6 favors t and $t + 3$; period 4 in base 8 favors t and $t + 4$, and patching it together with period 2 gives peaks at t and $t + 4$; period 5 in base 10 favors

t and $t + 5$; and period 2 favors the target's parity in bases 6, 8, and 10. Every harmonic of prime base 7, the fundamentals, and the full families have their highest value at the target. The base-10 parity write is the weakest (0.52 nats), and its alternation fades away from the target.

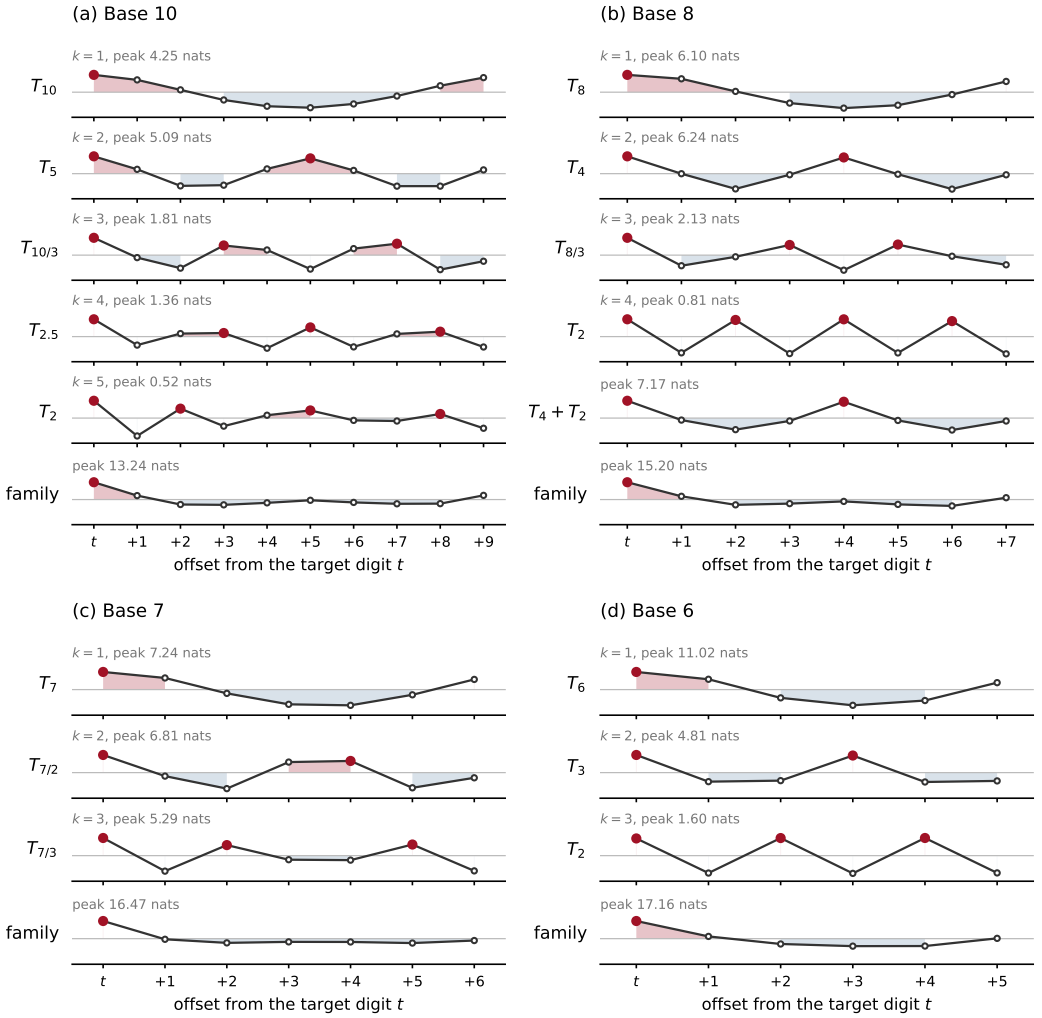


Figure 15: Harmonic contributions to digit scores for units digits after block 7 in bases 10, 8, 7, and 6, using the same cohorts and orthogonal planes as Figure 3 ($n = 1{,}500$ per base). Each row shows the change from the unedited digit logits, centered across digits within each case, aligned to the target digit t, and averaged over cases. Curves are divided by their own positive peak, with its size reported in nats; red points mark maxima. The family rows and the base-8 row for the joint patch of periods 4 and 2 show measured joint patches. The profiles show relative score changes; the final prediction also depends on the unedited scores (Appendix C.2).

C.3 Non-native Control Families

We fit equally sized control families on the frozen fitting rows and use identical evaluation pairs, ridge fits, centers, increasing-period orthogonalization, and phase edits (Appendix C.2). Table 5 lists the periods and full-family results. At the next digit, periods are scaled by the base and T_b is protected. Controls are orthogonalized within their own family; the native family is not projected out. Every control label has $|r| \leq 0.015$ with every native label on the frozen fitting and evaluation rows. Earlier period sets included periods with $|r| \approx 0.05$–0.06 on the fitting rows (for example T_{42} against T_{36} in base 6). A correlation of this size was enough for the ridge fit to return a plane close

Table 5: Layer-7 full-family steering on 1,500 pairs per setting. Δm is the mean target-minus-source margin gain in nats; follow is target following (%).

			Native		Control	
Base	Position	Control periods	Δm	Follow	Δm	Follow
6	Units	$\{9, 10, 15\}$	34.69	99.5	5.90	1.6
6	Next	$\{54, 60, 90\}$	33.39	97.3	1.90	0.3
7	Units	$\{3, 5, 10\}$	33.76	97.3	3.97	1.3
7	Next	$\{21, 35, 70\}$	33.17	96.6	2.70	0.3
8	Units	$\{3, 5, 6, 10\}$	31.31	94.7	4.09	0.6
8	Next	$\{24, 40, 48, 80\}$	31.60	98.0	2.97	0.4
10	Units	$\{3, 6, 12, 13, 15\}$	29.13	73.4	11.74	13.1
10	Next	$\{30, 60, 120, 130, 150\}$	28.62	78.7	7.81	5.9

to the corresponding native plane, because the native features dominate the activation covariance. These periods steered by up to 18.7 nats, and we replaced them.

No control plane carries information about its own period. A decoder fit on the fitting rows cannot recover the phase of $n \bmod T$ on held-out rows from any of the 30 orthogonal control planes ($R^2 \leq 0.015$; negative for 29 of them). Yet each plane has 4.6–42.1% of its energy in the native family, against 1.0–2.0% expected for a random plane in the 512-dimensional residual stream, and single-plane margin gain tracks this overlap across the 30 planes (Pearson $r = 0.80$, Spearman $\rho = 0.75$). Control steering is largest in base 10, where the mean overlap is 30.5%, against 8.0–14.9% in the other bases. The residual steering of the control families is therefore carried by native directions, not by the control periods. Single control planes give 0.3–3.0 nats and at most 1.1% target following. All self-patches reproduce the full logits exactly, and baseline digit logits match the native-family run.

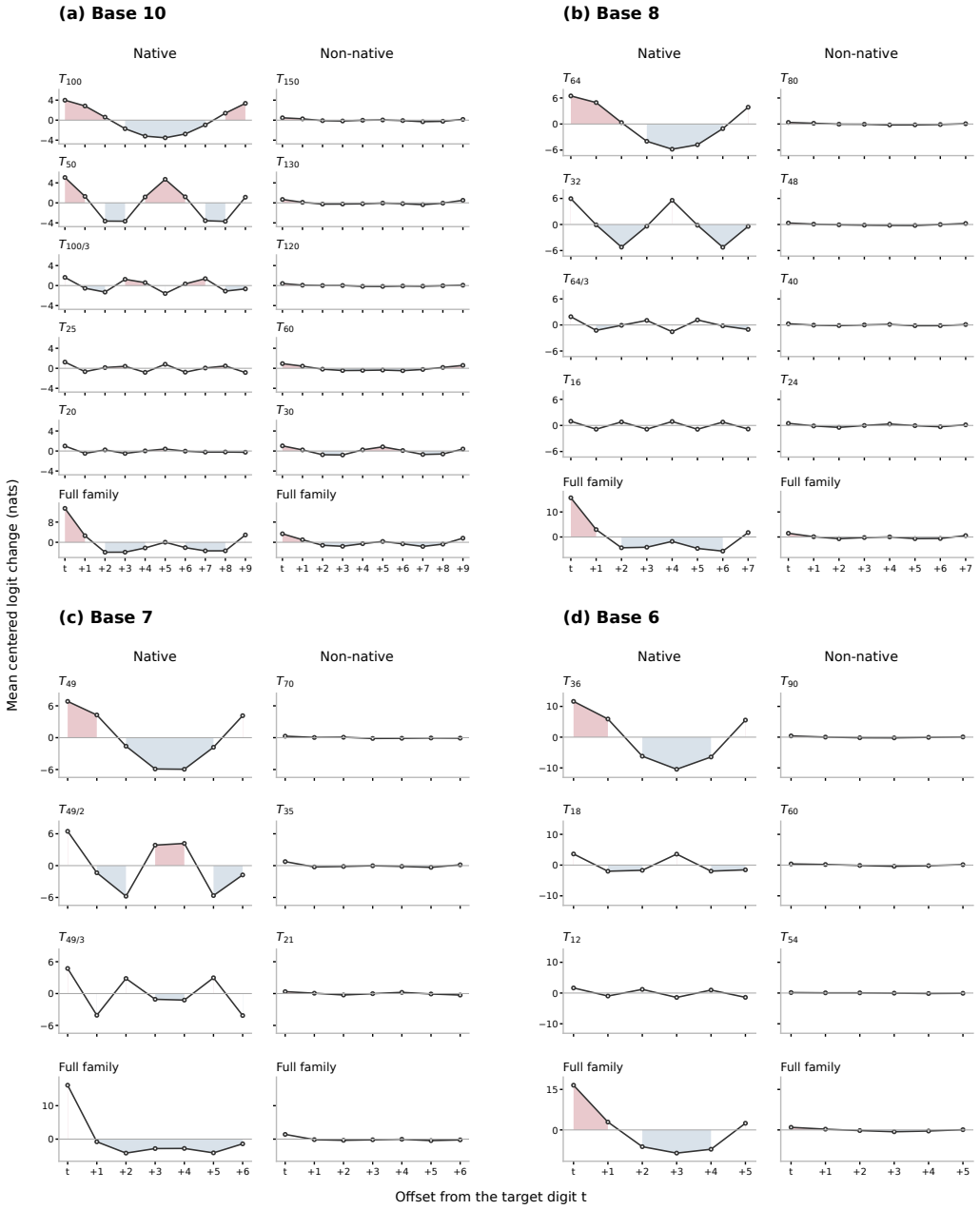


Figure 16: What native harmonics and non-native control periods write into the next-digit scores after block 7, on a common scale within each model (one scale for single periods and one for full families; $n = 1{,}500$ per model). Native rows are the harmonic family of each base; non-native rows are the control family of Table 5.

Period 2 in base 7. Base 7 has no period-2 harmonic, although the parity of a sum is well defined in any base and follows from the parities of the operands. We fit a period-2 plane at the units position with the integer's parity as label and patch it with the ideal parity sign, once as fitted and once orthogonal to the full native base-7 family. The fitted plane reaches 4.6 nats at layer 7 with at most 2.8% target following at any layer, the same leakage of native directions seen in the earlier control periods. The orthogonal plane changes no layer-7 prediction (0 of 1,500 target predictions, margin gain -0.010 nats, 95% interval $[-0.062, 0.043]$), and its largest effect at any layer is 1.6

nats at layer 4. Edits that use the donor's empirical sign give the same result. This bounds this intervention rather than every possible use of parity, and the next position was not tested.

C.4 Natural Contributions by Deletion

Patching turns each clock from the source digit to the target. To measure what the clocks contribute without a donor, we delete each harmonic on the same 1,500 source cases per setting, setting its centered coordinate to the fit intercept, $h' = h - \big((hQ) - c\big)Q^\top$, and leaving the other planes and the protected lower clock unchanged. Planes, intercepts, and cases are those of the patching runs, without refitting, and self-patches reproduce the full logits exactly.

At block 7, deleting the full family removes the model's preference for the correct digit. Before deletion, the correct digit leads the mean of the other digits by 16.4–17.5 nats. After deletion, the lead is between -1.2 and 1.1 nats in all eight settings, and accuracy falls by 77–89 percentage points.

Deletion moves the clock hand to the center, while patching turns it to the target (Figure 2d). On the target-minus-source margin, deletion therefore goes halfway, and patching should gain about twice as much, less where the target shares the source's stop. Measured patch gains are 0.89–1.00 times twice the deletion change for the full family and 0.89–1.45 times for single harmonics (Figure 3, open circles). Single harmonics exceed the prediction most in base 7. Deleting one plane can change the activation norm seen by the final normalization, and deleting the full family changes the margin by 1.09–1.43 times the sum of its single deletions, so single deletions likely understate the natural contribution.

C.5 Cross-Position Control

Patching a family at the other digit position tests whether each family acts only on its own digit. These arms come from an earlier, uncentered run of a two-position design in bases 7 and 8, after block 7 on 1,500 cases per setting. At each tested position we transplant the other position's family, fitted at that token without removing overlap with the native planes. At units the donors have the wrong phase, and at the upper position, where the lower periods take the same ideal phase for every suffix, the donors share the phase but change the digit. Table 6 gives the unions. Individual cross periods applied one at a time change at most 1 of 1,500 predictions in either base, although the unions edit three or four planes with half to two thirds of the native edit norm.

Table 6: Standalone cross-period unions at layer 7 in the earlier uncentered run. Rates are target following (%) on 1,500 cases.

Base	Tested position	Cross periods	Correct	Null
7	Units	$T_{49}, T_{49/2}, T_{49/3}$	0.00	0.00
7	Sevens	$T_7, T_{7/2}, T_{7/3}$	0.80	0.33
8	Units	$T_{64}, T_{32}, T_{64/3}, T_{16}$	0.00	0.00
8	Eights	$T_8, T_4, T_{8/3}, T_2$	0.00	0.00

D Additional Controls

D.1 Operand-Token Patching in Pythia

The patches in the main text act at the token before the answer digit, so the harmonics could in principle be built only for the output. To test whether they also carry the operands, we use Pythia-6.9B, which writes every number from 0 to 198 as a single token. The prompt is `Output ONLY a number. {a}+{b}=` with operands in 0–99, and the model predicts the whole sum as one token. We keep initially correct examples (7,963 of 10,000), fit on 4,000 of them, and evaluate on 1,500 disjoint cases per site. The units harmonics are fitted and orthogonalized with the protocol of Section 2, in the order $T_2, T_{5/2}, T_{10/3}, T_5, T_{10}$. The score of units digit d is the log-sum of the logits of all one-token answers with that units digit, so a difference of two scores is a log ratio of probability mass, in nats, and a patch follows the target when the decoded answer has the target units digit.

We first check the answer position. At the = token after block 26, the harmonics fitted to the sum behave as in the other models: patching the family with donors from the same decade moves the units digit in 86.7% of cases, the family gains 1.04 times the sum of the standalone gains, and each leave-one-out loss is within 0.1 nats of the standalone gain (Table 7).

We then patch the operands instead. At the tokens of a and b, we fit the harmonics of each operand's units digit separately for each operand and block, and write the coordinates of counterfactual operands into every block from 0 to 17 of the same forward pass. The answer's units digit then follows the sum of the counterfactual operands in 88.1% of cases. Writing each operand's own value changes almost nothing (0.4%), and writing a shuffled set of counterfactual operands gives 9.2%. The model therefore carries the operands' units digits in their harmonics and adds them, rather than only reading out a harmonic code at the answer. Because the patch acts at all 18 blocks together, it does not show that any single early block suffices.

Table 7: Pythia harmonic interventions ($n = 1{,}500$ per site). Each harmonic cell gives the standalone gain and the leave-one-out loss in nats; the last columns give the full-family gain and target following.

Site	T_{10}	T_5	$T_{10/3}$	$T_{5/2}$	T_2	Family	Following
Answer, block 26	1.9 / 2.0	2.7 / 2.8	0.55 / 0.57	0.87 / 0.92	2.4 / 2.5	8.7	86.7%
Operands, blocks 0–17	2.0 / 1.3	3.7 / 2.6	0.31 / 0.10	1.15 / 0.61	3.4 / 2.9	9.3	88.1%
own values						0.3	0.4%
shuffled operands						4.9	9.2%

D.2 Composing Angles from Two Donors

Our patches replace the angle of each plane with the donor's and keep the source's radius. The composition test takes the angles from two different donors, in the spirit of phase-swap experiments on images (Oppenheim & Lim, 1981; Yıldırım, 2026). For a source $a + b = c$ in Qwen after block 30, the period-2 angle comes from one donor sum and the period-5 angle from another, chosen so that the digit the two angles jointly specify appears in neither donor. For example, for a source sum ending in 4, the period-5 angle of a donor ending in 2 and the parity angle of a donor ending in 3 jointly specify the digit 7, which neither donor contains. The same is done at tens and hundreds with periods 20 and 50 and 200 and 500. The model moves to this digit, although it appears in no input, above norm-matched shuffled angles by 10.0, 16.3, and 13.0 pp at units, tens, and hundreds with the T_{10}-family plane left at its source value, and by 34.8, 35.3, and 32.3 pp after deleting it ($n = 800$ per condition; Figure 17). The residue angles are therefore read independently and combined downstream.

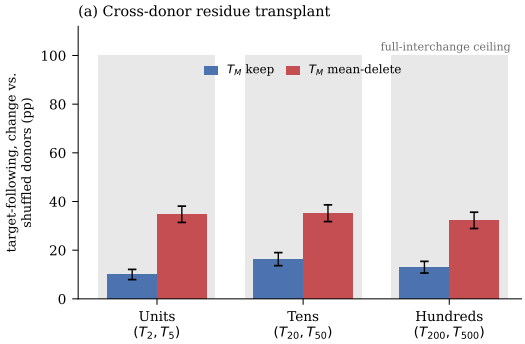


Figure 17: Composing angles from two donors in Qwen after block 30 ($n = 800$). Gain in target following over shuffled angles when the period-2 and period-5 angles come from different donors, so the target digit appears in no input.